Gaussian Mixture Models and the Expectation-Maximization Algorithm

1. Introduction to Gaussian Mixture Models

A Gaussian Mixture Model (GMM) is a probabilistic model used to represent a population as a mixture of several Gaussian distributions, each representing a different cluster or group. GMMs are commonly applied in unsupervised learning for clustering tasks, where each Gaussian component represents a distinct cluster.

Given a dataset $X = \{x_1, x_2, \ldots, x_N\}$, we assume that each data point x_i was generated by one of K Gaussian distributions. Each Gaussian component has:

- a mean μ_k,
- a covariance matrix Σ_k, and
- a mixing proportion π_k, which indicates the proportion of the data generated by component k.

The parameters of the GMM are denoted as $\theta = \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^{K}$.

Probability Density Function of a GMM

The probability of observing each data point x_i under the mixture model is:

$$p(x_i|\theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k),$$

where π_k is the mixing proportion for component k (with $\sum_{k=1}^{K} \pi_k = 1$), and $\mathcal{N}(x_i|\mu_k, \Sigma_k)$ is the Gaussian density function of component k:

$$\mathcal{N}(x_i|\mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2}|\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k)\right).$$

The Fundamental Challenge of Mixture Models

GMM estimation involves two coupled unknowns:

- **Parameter Uncertainty:** Unknown parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}$.
- **Assignment Uncertainty:** Unknown component labels $Z = \{z_1, \ldots, z_N\}$.

This creates a *chicken-and-egg problem*:

If we knew which points belonged to which clusters, parameter estimation would be simple. If we knew the parameters, assignments would be straightforward. But we know neither.

2. Using Latent Variables and the Complete Data Likelihood

To make the optimization problem more tractable, we introduce latent variables $Z = \{z_1, z_2, \ldots, z_N\}$, where each z_i is a hidden variable indicating the Gaussian component that generated x_i. Specifically, $z_i = k$ if x_i was generated by the k-th Gaussian component.

Given the latent variables Z, the likelihood of the complete data (X, Z) simplifies. The complete data likelihood assumes we know the specific component that generated each data point, and it is given by:

$$p(X, Z|\theta) = \prod_{i=1}^{N} \prod_{k=1}^{K} \left(\pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k)\right)^{\mathbf{1}(z_i=k)},$$

where $\mathbf{1}(z_i = k)$ is an indicator function that is 1 if $z_i = k$ (i.e., component k generated x_i) and 0 otherwise.

Complete Log-Likelihood

The complete log-likelihood is:

$$\log p(X, Z|\theta) = \sum_{i=1}^{N} \sum_{k=1}^{K} \mathbf{1}(z_i = k) \left(\log \pi_k + \log \mathcal{N}(x_i|\mu_k, \Sigma_k)\right).$$

This form is easier to work with because the log-sum is replaced by separate terms for each Gaussian component.

3. Expectation-Maximization (EM) Algorithm for GMMs

Since the latent variables Z are not observed, we cannot directly maximize the complete log-likelihood. Instead, we use the Expectation-Maximization (EM) algorithm, which iteratively maximizes the expected complete log-likelihood with respect to Z.

The EM algorithm alternates between two steps:

E-Step: Calculating Responsibilities

In the E-step, we calculate the expected value of the complete log-likelihood with respect to Z, given the observed data X and current parameter estimates $\theta^{(t)}$.

Calculating Responsibility (Posterior Probability)

Using Bayes' Rule, we can express this posterior probability as:

$$r_{ik} = p(z_i = k|x_i, \theta) = \frac{p(z_i = k)p(x_i|z_i = k, \theta)}{p(x_i|\theta)}.$$

Let's break down each term in this expression:

- $p(z_i = k)$: This is the prior probability of component k, which is just the mixing proportion π_k. So, $p(z_i = k) = \pi_k$.
- $p(x_i|z_i = k, \theta)$: This is the likelihood of observing x_i given that it was generated by component k. Since we assume each component follows a Gaussian distribution, this likelihood is given by:
$$p(x_i|z_i = k, \theta) = \mathcal{N}(x_i|\mu_k, \Sigma_k).$$
- $p(x_i|\theta)$: This is the marginal probability of observing x_i under the entire mixture model, taking into account all components. It is given by:
$$p(x_i|\theta) = \sum_{j=1}^{K} \pi_j \mathcal{N}(x_i|\mu_j, \Sigma_j).$$

Putting it all together, we have:

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_i|\mu_j, \Sigma_j)}.$$

This formula gives us the responsibility r_{ik} that component k takes for generating data point x_i. These responsibilities act as "soft" assignments of each data point to the components.

The expected complete log-likelihood is then given by:

$$Q(\theta|\theta^{(t)}) = \mathbb{E}_{Z|X,\theta^{(t)}} \left[\log p(X, Z|\theta)\right] = \sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \left(\log \pi_k + \log \mathcal{N}(x_i|\mu_k, \Sigma_k)\right).$$

Note that the probability $p(x_i|\theta)$ represents the marginal likelihood of observing data point x_i, obtained by integrating over all possible latent component assignments. This follows directly from the law of total probability:

$$p(x_i|\theta) = \sum_{k=1}^{K} p(x_i, z_i = k|\theta) = \sum_{k=1}^{K} p(z_i = k|\theta) \cdot p(x_i|z_i = k, \theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k),$$

where we marginalize over the latent variable z_i, representing which component generated x_i.

Log-Likelihood of the GMM

Given N independent observations $X = \{x_1, x_2, \ldots, x_N\}$, the log-likelihood for the entire dataset is:

$$\log p(X|\theta) = \sum_{i=1}^{N} \log \left(\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k) \right).$$

Maximizing this log-likelihood directly is challenging because it involves a log of a sum over components, which complicates parameter optimization.

M-Step: Maximizing the Expected Complete Log-Likelihood

In the M-step, we maximize $Q(\theta|\theta^{(t)})$ with respect to the parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}$ to obtain updated parameter estimates $\theta^{(t+1)}$.

1. Maximizing with Respect to π_k: Mixing Proportion

The mixing proportions π_k represent the probability of each component in the mixture and must satisfy the constraint:

$$\sum_{k=1}^{K} \pi_k = 1 \quad \text{and} \quad \pi_k \geq 0.$$

Setting Up the Lagrangian To incorporate the constraint $\sum_{k=1}^{K} \pi_k = 1$, we introduce a Lagrange multiplier λ and set up the following Lagrangian:

$$\mathcal{L}(\pi_k, \lambda) = \sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \log \pi_k + \lambda \left(1 - \sum_{k=1}^{K} \pi_k \right).$$

Taking the Derivative with Respect to π_k To find the maximum, we take the derivative of $\mathcal{L}$ with respect to π_k and set it to zero:

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{\sum_{i=1}^{N} r_{ik}}{\pi_k} - \lambda = 0.$$

Solving for π_k:

$$\pi_k = \frac{\sum_{i=1}^{N} r_{ik}}{\lambda}.$$

Using the Constraint To determine λ, we use the constraint $\sum_{k=1}^{K} \pi_k = 1$:

$$\sum_{k=1}^{K} \pi_k = \sum_{k=1}^{K} \frac{\sum_{i=1}^{N} r_{ik}}{\lambda} = \frac{1}{\lambda} \sum_{k=1}^{K} \sum_{i=1}^{N} r_{ik} = 1.$$

Solving for λ, we find $\lambda = N$, giving:

$$\pi_k = \frac{1}{N} \sum_{i=1}^{N} r_{ik}.$$

This is the updated formula for π_k, which represents the average responsibility that component k takes for all data points.

2. Maximizing with Respect to μ_k: Mean

For the mean μ_k, we want to maximize:

$$Q(\mu_k) = \sum_{i=1}^{N} r_{ik} \log \mathcal{N}(x_i|\mu_k, \Sigma_k).$$

Expanding $\log \mathcal{N}(x_i|\mu_k, \Sigma_k)$ (dropping constants that do not depend on μ_k):

$$Q(\mu_k) = -\frac{1}{2} \sum_{i=1}^{N} r_{ik} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k).$$

Taking the Derivative with Respect to μ_k We take the derivative of $Q(\mu_k)$ with respect to μ_k and set it to zero:

$$\frac{\partial Q(\mu_k)}{\partial \mu_k} = \sum_{i=1}^{N} r_{ik} \Sigma_k^{-1} (x_i - \mu_k) = 0.$$

Rearranging terms:

$$\sum_{i=1}^{N} r_{ik} x_i = \mu_k \sum_{i=1}^{N} r_{ik}.$$

Solving for μ_k:

$$\mu_k = \frac{\sum_{i=1}^{N} r_{ik} x_i}{\sum_{i=1}^{N} r_{ik}}.$$

This is the updated formula for μ_k, which is a weighted average of the data points, where each point is weighted by its responsibility r_{ik} for component k.

3. Maximizing with Respect to Σ_k: Covariance Matrix

For the covariance matrix Σ_k, we want to maximize:

$$Q(\Sigma_k) = \sum_{i=1}^{N} r_{ik} \log \mathcal{N}(x_i|\mu_k, \Sigma_k).$$

Expanding $\log \mathcal{N}(x_i|\mu_k, \Sigma_k)$ (dropping constants that do not depend on Σ_k):

$$Q(\Sigma_k) = -\frac{1}{2} \sum_{i=1}^{N} r_{ik} \left(\log |\Sigma_k| + (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right).$$

Taking the Derivative with Respect to Σ_k To find the maximum, we take the derivative of $Q(\Sigma_k)$ with respect to Σ_k and set it to zero. Using matrix calculus, we obtain:

$$\frac{\partial Q(\Sigma_k)}{\partial \Sigma_k} = -\frac{1}{2} \sum_{i=1}^{N} r_{ik} \left(\Sigma_k^{-1} - \Sigma_k^{-1} (x_i - \mu_k)(x_i - \mu_k)^T \Sigma_k^{-1} \right) = 0.$$

Rearranging, we get:

$$\Sigma_k = \frac{\sum_{i=1}^{N} r_{ik} (x_i - \mu_k)(x_i - \mu_k)^T}{\sum_{i=1}^{N} r_{ik}}.$$

This is the updated formula for Σ_k, which is the weighted covariance of the data points assigned to component k.

Summary of EM Steps

1. **E-Step**: Calculate responsibilities $r_{ik} = p(z_i = k|x_i, \theta^{(t)})$ using current parameters $\theta^{(t)}$:

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_i|\mu_j, \Sigma_j)}.$$

2. **M-Step**: Update parameters using the responsibilities:

$$\pi_k = \frac{1}{N}\sum_{i=1}^{N} r_{ik}, \quad \mu_k = \frac{\sum_{i=1}^{N} r_{ik} x_i}{\sum_{i=1}^{N} r_{ik}}, \quad \Sigma_k = \frac{\sum_{i=1}^{N} r_{ik}(x_i - \mu_k)(x_i - \mu_k)^T}{\sum_{i=1}^{N} r_{ik}}.$$

By iteratively applying these steps, the EM algorithm maximizes the marginal log-likelihood $\log p(X|\theta)$ with respect to the parameters, yielding the optimal parameter estimates for the GMM.

Initialization and Convergence

The EM algorithm requires initial parameter values $\theta^{(0)}$. Common initialization strategies include:

- Random initialization of responsibilities
- K-means clustering for initial component assignments
- Random selection of data points as initial means

The algorithm typically converges when the change in log-likelihood between iterations falls below a predefined threshold:

$$|\log p(X|\theta^{(t+1)}) - \log p(X|\theta^{(t)})| < \epsilon.$$

Note on Covariance Estimation In practice, regularization may be applied to prevent singular covariance matrices:

$$\Sigma_k = \frac{\sum_{i=1}^{N} r_{ik}(x_i - \mu_k)(x_i - \mu_k)^T}{\sum_{i=1}^{N} r_{ik}} + \epsilon I,$$

where ϵ is a small positive constant and I is the identity matrix.

Why $\mathbb{E}_{Z|X,\theta^{(t)}}\Big[\log p(X, Z \mid \theta)\Big]$ takes the familiar GMM form

Setup. For a Gaussian mixture model with K components and parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^{K}$, introduce latent labels $Z = \{z_1, \ldots, z_N\}$ where each $z_i \in \{1, \ldots, K\}$ indicates the component that generated x_i. Equivalently, use one-hot indicators $z_{ik} \in \{0, 1\}$ with $\sum_{k=1}^{K} z_{ik} = 1$ and $z_{ik} = \mathbf{1}(z_i = k)$.

Complete-data likelihood and log-likelihood. By conditional independence given Z,

$$p(X, Z \mid \theta) = \prod_{i=1}^{N} \prod_{k=1}^{K} \Big(\pi_k \, \mathcal{N}(x_i \mid \mu_k, \Sigma_k)\Big)^{z_{ik}}.$$

Taking logs,

$$\log p(X, Z \mid \theta) = \sum_{i=1}^{N} \sum_{k=1}^{K} z_{ik} \Big(\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \Sigma_k)\Big). \tag{$*$}$$

Posterior over labels factorizes across i. Given parameters $\theta^{(t)}$, the posterior over labels factorizes:

$$p(Z \mid X, \theta^{(t)}) = \prod_{i=1}^{N} p(z_i \mid x_i, \theta^{(t)}),$$

and each z_i is categorical with

$$r_{ik} \;\; := \;\; p(z_i = k \mid x_i, \theta^{(t)}) = \frac{\pi_k^{(t)} \, \mathcal{N}(x_i \mid \mu_k^{(t)}, \Sigma_k^{(t)})}{\sum_{j=1}^{K} \pi_j^{(t)} \, \mathcal{N}(x_i \mid \mu_j^{(t)}, \Sigma_j^{(t)})}.$$

Key expectation. Because $z_{ik} \in \{0, 1\}$ and z_i is categorical with probabilities $r_{i1}, \ldots, r_{iK}$,

$$\mathbb{E}_{Z \mid X, \theta^{(t)}}[z_{ik}] = p(z_i = k \mid x_i, \theta^{(t)}) = r_{ik}.$$

By linearity of expectation, apply this to each summand in $(*)$:

$$\mathbb{E}_{Z \mid X, \theta^{(t)}}\big[\log p(X, Z \mid \theta)\big] = \sum_{i=1}^{N} \sum_{k=1}^{K} \mathbb{E}[z_{ik}] \Big(\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \Sigma_k)\Big).$$

Substitute $\mathbb{E}[z_{ik}] = r_{ik}$ to obtain

$$\boxed{\mathbb{E}_{Z \mid X, \theta^{(t)}}\big[\log p(X, Z \mid \theta)\big] = \sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \Big(\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \Sigma_k)\Big).}$$

Remarks.

- This expected complete-data log-likelihood is the Q-function in EM: $Q(\theta \mid \theta^{(t)}) := \mathbb{E}_{Z \mid X, \theta^{(t)}}[\log p(X, Z \mid \theta)]$.
- The E-step computes r_{ik} (posterior responsibilities). The M-step maximizes the boxed expression over θ, yielding the standard weighted MLE updates for π_k, μ_k, Σ_k.